Progressive Token Reduction and Compensation for Hyperspectral Image Representation

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***Abstract*— Hyperspectral images (HSIs) have been widely used in Earth observation because they contain continuous and detailed spectral information which is beneficial for the fine-grained diagnosis of the land cover. In the past few years, convolutional neural network (CNN)-based methods show limitations in modeling spectral-wise long-range dependences. Recently, transformer-based deep learning methods are proposed and have shown superiority in modeling the continuous representation of the spectral signatures because the self-attention (SA) mechanism has a global receptive field. Due to the special tokenization of the transformer-based methods, the redundant tokens contained in spectral embeddings are always involved in SA operation. Redundant tokens do not positively contribute to classification. Specifically, the overlapped group-wise tokenization approach may aggravate the Hughes phenomenon and impose additional computations. To address this issue, a lightweight spatial–spectral pyramid transformer (SSPT) framework is proposed to efficiently extract the spatial–spectral features of HSI by progressively reducing redundant tokens in an end-to-end manner. In particular, a token reduction (TR) method is proposed to decide which tokens will be involved by computing and comparing token attentiveness between spectral embeddings and the class token. In addition, for those tokens that are defined as redundant information, a token compensation mechanism is proposed to automatically extract supplementary information for classification. Extensive experiments on three standard datasets quantitatively show the superiority of our methods, and the ablation experiments qualitatively prove our hypothesis about the feature distribution in transformer architecture.**

***Index Terms*— Hyperspectral image (HSI) classification, redundant information, token compensation, token reduction (TR), transformer network.**

I. Introduction

HYPERSPECTRAL satellites have developed rapidly in recent years. ZY1-02D, ZY1-02E, GF-5, and other satellites have been launched one after another which provides massive data for the application of precision agriculture [1], environmental monitoring [2], mineral exploration [3], and other natural resource exploration. The diagnosis of land cover is fundamental to the application of hyperspectral data, and its reliability and refinement are crucial to Earth observation tasks.

The implementation of fine-grained classification relies on the extraction of reliable features of HSIs. Traditional machine learning methods extract hand-crafted features and use conventional classifiers, such as support vector machines (SVM) [4], random forests (RF) [5], or K-nearest neighbor (KNN) [6] for classification. To improve the spatial–spectral representation ability of hand-crafted features, some kernel-based learning methods (e.g., RBF kernel [7] and morphological kernel [8]) are introduced to generate reliable spatial–spectral representations of HSIs. The hand-crafted spatial–spectral-based methods obtain good classification results and perform well with limited training samples. However, these methods are easy to encounter bottlenecks when faced with large amounts of complex remote sensing datasets.

Recently, deep learning (DL) architectures have been successfully applied in processing vast high-dimensional remote sensing data, because the DL methods can extract potentially effective features from massive data by integrating different backbone networks. The common deep learning architectures including stacked auto-encoder (SAE) [9], deep belief network (DBN) [10], CNN [11], [12], [13], [14], recurrent neural networks (RNNs) [15], GAN [16], and graph convolutional network (GCN) [17], [18] have a mature application in HSI processing. For example, Feng et al. [19] proposed a GAN-based self-supervised divide-and-conquer network (SDC-GEN), which introduced self-supervised learning into GAN for HSI classification. In this method, a pretext cluster task was proposed to precisely divide limited labeled samples with the assistance of abundant unlabeled samples. A 3-D CNN-based spatial–spectral residual network was proposed by Zhong et al. [20] for HSI classification. Xu et al. [21] used a dual-branch structure composed of 1-D and 2-D CNN to extract spectral and spatial features, and then the fully connected layers were used to fuse features. Wang et al. [22] introduced the GhostNet based on 2-D CNN to extract spatial features. Gao et al. [23] proposed a multiscale residual architecture with the idea of depth-wise convolution which reduced the computational cost and maintained promising classification accuracy. Feng et al. [24] proposed a network named bandwise-independent and hard thresholding convolutional neural network (BHCNN) that integrates HSI band selection and classification. However, due to the fixed

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kernel size and limited receptive field of CNNs, CNN-based methods show limitations in modeling long-range interspectra dependences. The hierarchical architecture of CNNs distorts the continuous attributes of spectra because it overly focuses on locally spatial information.

Very recently, transformer network [25] is proposed to model the sequence data in the field of natural language processing (NLP). Different from other deep learning models, the transformer has a global receptive field even in the shallow layer. Vision transformer (ViT) [26] first applies SA mechanism to the image classification task that turns the image into sequence data by a linear projection layer. Inspired by ViT model, some researchers directly apply the transformer to HSI classification. For example, He et al. [27] proposed a BERT-based hyperspectral image (HSI) classification framework and used joint training to adopt different regions of target pixels. Qing et al. [28] divided the entire HSI into non-overlapping patches in the spatial dimension and then adopted the spectral attention module before the SA mechanism, which consisted of pooling operations.

To integrate the merits of both CNNs and transformer, many joint CNN and transformer networks are proposed. For example, Hong et al. [29] rethought the application of transformer and proposed SpectralFormer (SF) model which generated group-wise spectral embeddings from neighboring spectral bands. In [30], [31], [32], and [33], various CNN-based spatial–spectral feature extraction networks had been proposed to generate more representative tokens for transformer architecture. In [34], a transformer-based integrated reconstruction and classification architecture named masked auto-encoding spectral-spatial transformer (MAEST) was proposed. Yang et al. [35] proposed a new HSI transformer (HiT) that embedded CNNs into the transformer. Meanwhile, other successful architectures in computer vision that embedded convolution into transformer (i.e., Deep ViT [36], T2T [37], LeViT [38], and RvT [39]) had also been applied to demonstrate the effectiveness of HiT.

The aforementioned methods use the ViT-like backbones (we called transformer-based model) to learn the spectral-wise long-term dependences that achieve better performance than their counterparts in fine-grained classification. However, the great number of channels of HS data cube causes the curse of dimensionality [40]. The redundant information between spectral bands of high-dimensional data is inevitably involved in spectral embeddings which makes the network difficult to extract effective features and brings an additional computational burden. To be specific, the number of parameters and computational volume [determined by floating-point operations per second (FLOPS)] in the transformer is proportional to the depth and a quadratic function of the width [41]. As depicted in Fig. 1, where (h, w) represents the spatial resolution of the image patch, n, g, and d represent the number of spectral bands, group size, and embedded dimension, respectively. Each token generated by band-wise or group-wise tokenization represents an embedding of one spectral channel or a group of neighboring spectral channels. During the tokenization stage, the redundant information between spectral bands becomes redundant tokens contained in

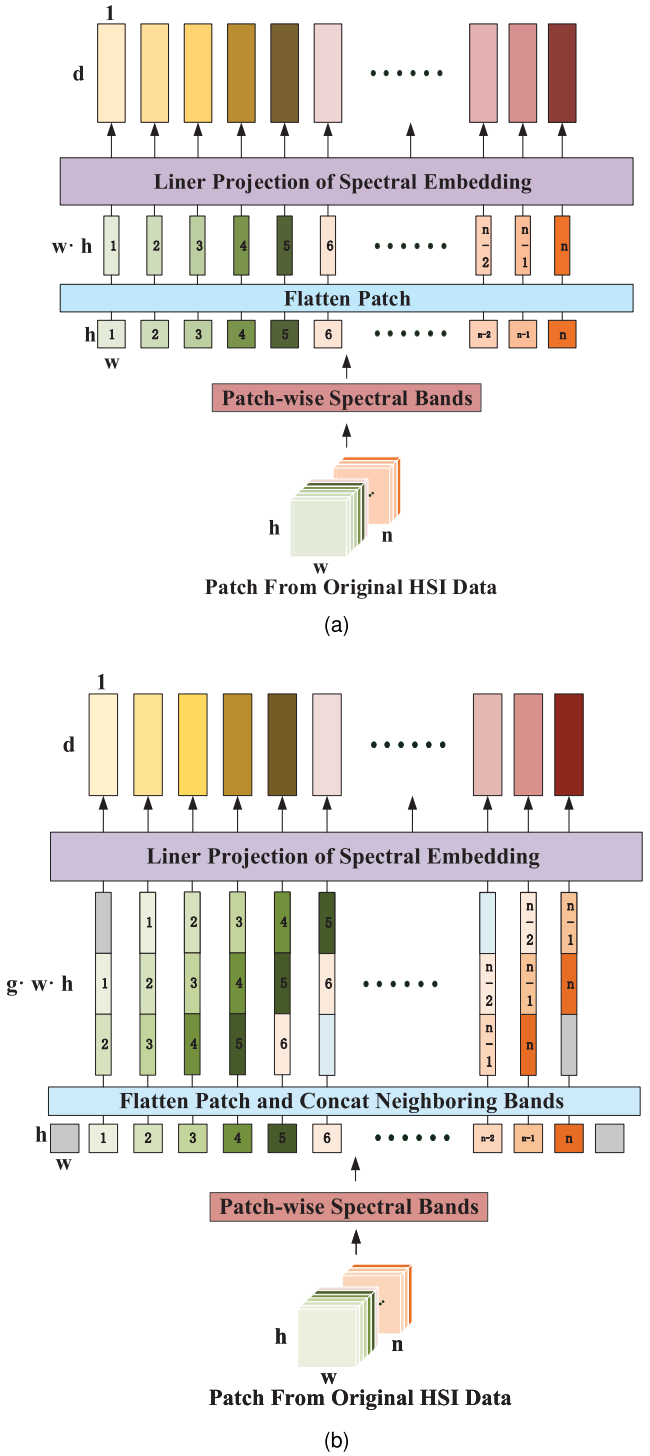

Fig. 1. Illustration of different tokenization methods of HSI. (a) Band-wise. (b) Group-wise. (The gray rectangles represent zero padding.)

spectral embeddings. Meanwhile, the working dimension of the transformer-based methods remains unchanged after tokenization, which leads to redundant information being continuously computed inside the transformer. Comparing Fig. 1(a) and (b) (take group size $g = 3$ as an example), the increase in group size not only brings more spatial–spectral information but also increases redundant information, which may lead to a decline in classification performance and increase the computational burden. We conducted band-wise ($g = 1$) and group-wise ($g > 1$) tokenization for ViT with

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TABLE I
SENSITIVE ANALYSIS OF THE NUMBER OF NEIGHBORING BANDS IN VIT MODEL IN TERMS OF OA (%) ON IP DATASET

Methods	Input	Group Size (g)				
		1	3	5	7	9
ViT	pixel-wise	72.36	74.83	**75.86**	74.67	74.09
	patch-wise	80.08	82.28	**83.88**	83.64	83.65

pixel-wise ($h = w = 1$) and patch-wise ($h = w = 3$) inputs, respectively, to illustrate the degradation of classification performance caused by too many redundant features. The results in Table I show that the increase in the input information (expand group size) does not lead to a continuous increase in classification performance, even if we adopted some drop mechanisms [42], [43] or regularization methods to prevent over-fitting. We deem that redundant features are introduced along with valid features, which will cause semantic ambiguity and reduce classification accuracy.

To address the aforementioned limitations and inspired by some efficient design of transformer [44], [45], [46], an end-to-end SSPT network is proposed, which takes the advantage of the SA mechanism to identify the redundant tokens without losing the fine features of spectral attributes. First, given all bands of one patch from HSI as input, we adopt group-wise tokenization to generate spectral embeddings. This operation directly increases locally detailed discrepancies of spectra. Second, the contribution of each spectral embedding to the class token is computed by SA mechanism and some tokens with higher attention scores are selected directly as the input for the next stage. In this way, the redundant features are progressively reduced as the network goes deeper. Third, we propose an adaptive token compensation (ATC) mechanism that can extract supplementary information from redundant features in every reduction stage. Finally, the classifier of ViT is used to generate the classification result.

The main contributions of this article are summarized as follows.

1) A lightweight and pure transformer-based HSI classification framework named SSPT is proposed in this article. The architecture does not contain any hand-crafted preprocessing and convolution operations.
2) A token reduction (TR) method is proposed in our SSPT network for distinguishing valid and redundant information among spectral bands. This reduction method improves the classification performance while significantly reducing the computational burden of the model.
3) Different from the existing straightforward fusing strategy, an ATC mechanism is developed, which is able to extract supplementary information from unselected spectral embeddings in every encoder stage.
4) All the methods proposed in SSPT can be easily plugged into the transformer-based structure which is applied for downstream applications of HSI data. The experimental results demonstrate that the proposed SSPT can significantly reduce the computational cost of transformer-based methods while improving classification performance. Meanwhile, it maintains superiority in fine-grained classification over other state-of-the-art backbone networks.

The rest of this article is structured as follows. Section II introduces the basic concepts of ViT and related works. Section III describes the proposed method in detail. The experimental results on three HSI datasets with ablation studies are shown in Section IV. Finally, we summarize some conclusions in Section V.

II. RELATED WORKS

In this section, we introduce some preliminaries of ViT and some methods to lighten the transformer by changing the network structure and attention operations.

A. Formulation of ViT

ViT first applies the transformer's design to vision tasks. To well keeping the original architecture of transformer, the 2-D input image $X \in \mathbb{R}^{w\times h\times c}$ is reshaped into image sequence $X \in \mathbb{R}^{n\times(p\cdot c^2)}$ by convolution and flatten operation, where (w, h) is the spatial resolution of the original image, and $n = hw/p^2$ is the sequence length determined by the patch size p. Because the ViT uses constant widths in all of its layers, the final input of the transformer encoder is a linear mapping of each sequence.

Very recently, ViT and its variants have achieved state-of-the-art effects on various vision tasks. The success of ViT depends much on the stacked with several transformer blocks of the same architecture which can well model the long-range dependence even in the shallow layer. In ViT architecture, each encoder block is composed of a multihead attention layer (MHSA), a multilayer perceptron (MLP) layer, shortcut connection, and layer normalization (LN) [47].

In SA layer, the input token vector is first linearly transformed into query (q) vector, key (k) vector, and value (v) vector, and then all of the q, k, v vectors form three matrices, named Q, K, and V. Third, the attention map A is computed by the inner product Q and K matrices and normalizes the scores for the stability of gradient by dividing $\sqrt{d_k}$, where d_k is the dimension of k vector. Fourth is applying the softmax function to the attention score. At last, the attention representation is obtained by time V matrix. The whole SA operation can be formulated in (1). To increase the diversity of features in the actual vision tasks, MHSA is used to boost the performance of the vanilla SA layer, and the details of MHSA are formulated as follows:

$$\mathrm{SA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V = A \cdot V \tag{1}$$

$$\mathrm{MHSA}(Q, K, V) = \mathrm{Concat}(\mathrm{SA}_1, \mathrm{SA}_2, \ldots, \mathrm{SA}_h)W \tag{2}$$

where h and W are the numbers of head and projection weight, respectively.

The MLP layer contains two linear transformation layers and a nonlinear activation function, and it can be denoted as the following function:

$$\mathrm{MLP}(X) = W_2\sigma(W_1X) \tag{3}$$

where W_1 and W_2 are the parameters of the linear layer and σ is Gaussian error linear units (GELU) [48]. In addition,

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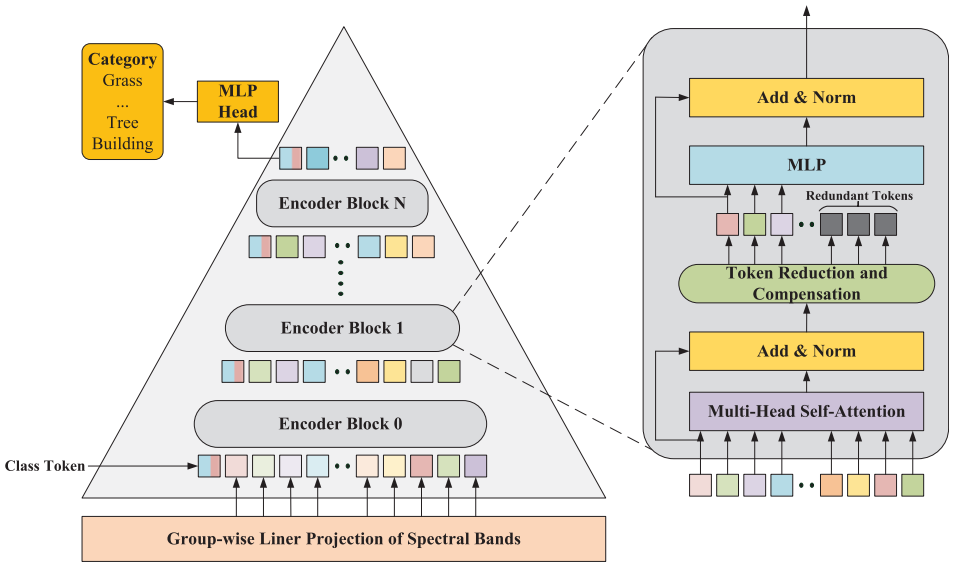

Fig. 2. Overview illustration of the proposed SSPT network for the HSI classification. Each encoder block of SSPT contains a TR and compensation module.

the shortcut connection is contained in each sub-layer of the encoder block and it can be described as follows:

$$X = \mathrm{LN}(X + S(X)) \tag{4}$$

where $S(\cdot)$ can be either MHSA or MLP operation.

B. Lightweight Design of Transformer Structure

As depicted in Section II-A, the SA mechanism is the most critical part of the transformer. The mechanism enables each token to learn information from other tokens under the sequential input that improve the network's ability in modeling long-term dependences. However, the quadratic cost with input sequence length of the SA mechanism impacts the overall computation cost of the ViT [49]. To alleviate the contradiction between the network performance and the time memory complexity that exists in the transformer-based architecture, some efficient and lightweight models are designed to find a solution to the contradiction.

For example, Wang et al. [50] applied the idea of low-rank projections into SA which projected the length dimension into low-dimensional space. Kitaev et al. [51] used locality-sensitive hashing (LSH) instead of dot-product attention, which made the computational complexity drop directly from $O(L^2)$ to $O(L\log L)$, where L is the sequence length. A linear transformer [52] was proposed to reduce the complexity from quadratic to linear by kernel-based formulation, and the causal masking further shortened the computation time of attention. Ryoo et al. [53] proposed a learnable token generation module named TokenLearner which made the subsequent layers only to process a small number of tokens. In summary, the aforementioned methods focus on reducing the calculated dimension of the SA mechanism by changing the way attention is computed or adding additional modules.

III. Materials and Methods

In this section, we propose the SSPT network. Fig. 2 shows the macro-structure of SSPT, and the tokenization method we adopted is depicted in Fig. 1(b). The solid-colored squares in Fig. 2 represent the spectral embeddings with learnable location information, and the mixed-colored square represents an extra class token. Similar to ViT structure, SSPT consists of a series of encoders with the same components. Compared to the encoder in ViT, the encoder of SSPT has an additional TR and compensation module which will be described in detail later. Note that the gray squares in the encoder are tokens that are judged to contain redundant information and will be discarded from subsequent operations.

A. Token Reduction

Before the classifier, the working dimension (number of calculated tokens) is constant in ViT. If a similar structure is used directly to model the long-range interspectra dependences, tokens that contain redundant information are always involved in the feature extraction stage and continuously interact information with the class token because of the special calculation model of the SA mechanism. Intuitively, as depicted in Fig. 1, the valid information (both spatial and spectral dimensions) contained in the spectral embeddings will continuously increase as the number of input group size increase, which provides more detailed information that helps to further improve the classification performance of the deep learning methods. However, according to our experimental results, the increase in group size does not bring a sustained gain in classification accuracy, and even appears to be counterproductive. We deem that as the information contained in spectral embedding increases, the percentage of redundant information will also increase which is not only computation unfriendly but also weakens the network performance to efficiently learn the effective features, especially under the limited training samples. Although some methods use image preprocessing algorithms like principal component analysis (PCA) [31], [54] to successfully remove redundant and noisy bands from the original HSI data, this non-end-to-end processing is not conducive to the deployment of edge devices. Our proposed TR method progressively removes redundant information in an end-to-end and non-parametric manner to effectively model

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the long-range interspectra dependences and generate reliable representations of the HSI data.

Reviewing the attention operation of transformer in (1), the matrix A is named attention map. Each row of the attention map represents the attention score from the current token to all tokens, and the score indicates the dependence between the current token and all tokens. For the classification task, the class token is separated from embedding after the last encoder block, and the interactions between class token and other tokens are calculated as follows:

$$x_{\text{class}} = \text{softmax}\left(\frac{q_{\text{class}} \cdot K^T}{\sqrt{d_k}}\right) V = \alpha \cdot V \tag{5}$$

where q_{class}, K, and V mean the query vector of the class token, the key matrix, and the value matrix, respectively. α denotes the first row of the attention map A and will be used to represent the contribution of each token for classification. Since the key matrix and value matrix contain all the tokens, we can assume that each element of α determined the importance of the ith token. For those tokens with high attention scores, Liang et al. [44] defined them as attentive tokens and the remainder are defined as inattentive tokens. Since ViT divides the optical images into a series of non-overlapped patches in the spatial dimension, attentive tokens represent the valid features of targets and inattentive tokens contain background information outside the targets. However, we further divide the original HSI patches in the channel dimension which makes the SA mechanism to directly capture the correlation between spectral bands. Based on these operations, we deem that the inattentive tokens may represent redundant information or noise bands which do not positively contribute to classification. In order to identify attentive tokens that are valuable for the classification task, the top-K algorithm is used to extract the top K tokens with high attention scores (α). The number of attentive tokens K is calculated in (6), where γ and i represent the keep rate and the index of the encoder block, respectively. The initial K (K_0) depends on the number of bands of the input data

$$K_i = \gamma * K_{i-1}. \tag{6}$$

For MHSA operation, all the heads of each token will perform the attention operation at the same time and each head will have an attention vector. We compute the average attention score of all heads in (7), where N and n indicate the total number of attention heads and index of heads, respectively. The result of the average attention score (α_i) represents the attention score of ith token. Finally, the selected tokens will participate in the subsequent feature extraction stage

$$\alpha_i = \Sigma_{n=1}^{N} \frac{\alpha^n}{N}. \tag{7}$$

B. ATC Mechanism

The inattentive tokens in ViT model represent much meaningless background information outside the target. However, the tokens on the backgrounds may still be able to contribute to the classification tasks (i.e., birds are more likely to be in the sky background than fish). Based on this analysis, Liang et al. [44] proposed an inattentive token fusion (TF) strategy that weighted the average of the inattentive tokens as a new supplementary token. The fuse operation can be written as (8), where N is the index set of the inattentive tokens and α_i represents the attention score of corresponding inattentive tokens

$$x_{\text{fused}} = \sum_{i \in N} \alpha_i x_i. \tag{8}$$

The fusion strategy mentioned earlier, however, may differ from our hypothesis; the contribution of inattentive tokens of HSI to classification may be very limited or even negative in the early training stage; instead, the inattentive token in the late training stage may contain part of the deep semantic information that are positively contributing to classification. Our experimental results shown in Table II will further support our hypothesis (each stage represents the index of the encoder block and $\surd$, $\times$ represents whether to adopt a fusion strategy in the current encoder).

When we directly adopt the fusion strategy in (8) for all the feature extraction stages, the OA score is even lower than bluntly discarded inattentive tokens. The reason for this phenomenon is that we adopt a different way of dividing the image data than ViT. In detail, the ViT divides images into non-overlapping tokens in the spatial dimension; however, we divide the HSI patch into a series of tokens in the channel dimension. Moreover, if the group size is larger than 1, the spatial–spectral features between tokens will overlap. As we described in Section I, the group-wise tokenization method inevitably introduces more redundant information while increasing the valid information. So directly fusing the inattentive tokens for all stages may reintroduce redundant features, which ultimately leads to degradation of classification performance. In addition, the bluntly discarded operation may be effective in the shallow layer of the network, because most of the redundant information is contained in the original input of spectral embeddings. However, subsequent inattentive tokens (selected from the preceding attentive ones) may contain more valid semantic information than the preceding inattentive tokens. To further prove our hypothesis, we adopt the fusion strategy in different stages. The experimental results in Table II show that the inattentive tokens at different stages contribute to the classification differently. In detail, comparing case 0 with others, when only one stage adopts the fusion strategy, the shallow layer fuses more inattentive tokens but does not lead to higher classification accuracy. Then, comparing case 1 and case 2 with case 3, the inattentive tokens in the intermediate stage bring relatively high gains for classification, which may be that the number of inattentive tokens containing deep semantic information in the final stage is much smaller than the middle ones.

Based on the above-mentioned analysis, we deem that the more inattentive tokens in the shallow layer should be dropped and the proportion of inattentive tokens in x_{fused} should gradually increase as the network goes deeper. To select relatively effective features from inattentive tokens as much as possible, we propose an ATC mechanism to decide the proportion of the inattentive tokens that need to be fused as supplement

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TABLE II
ANALYSIS OF THE EFFECT OF INATTENTIVE TOKENS IN TERMS OF OA (%) AT DIFFERENT STAGES ON THE IP DATASET

Method	Input	Encoder Block				OA (%)
		stage0	stage1	stage2	stage3	
ViT+TI	Pixel-wise	×	×	×	×	73.12
ViT+TF		√	√	√	√	72.89(↓ 0.23)
case0		√	×	×	×	73.24(↑ 0.12)
case1		×	√	×	×	**73.60**(↑ 0.48)
case2		×	×	√	×	73.49(↑ 0.37)
case3		×	×	×	√	73.29(↑ 0.17)

ones. Concretely, for each stage of SSPT, we calculate the qth quantile (Q) of the attention score of inattentive tokens (α_{inatten}) as follows:

$$Q_i = \text{torch.quantile}(\alpha_{\text{inatten}_i}, q_i) \tag{9}$$

where i represents the index of the encoder block and the quantile Q is selected as the fusion threshold to distinguish the tokens that need to be fused and dropped. Those inattentive tokens whose attention score is higher than the fusion threshold will adopt the fusion strategy as (8). q is generated by a function 10, where i is the index of stage and $\beta \in [0, 1)$ is hyperparameter that determines the distribution of q. Since the number of inattentive tokens in the deep network is relatively small, we use the $\sin(\cdot)$ function to make the fusion threshold change more slowly at the later stage without changing the distribution trend of q. To further prove that the information in tokens obeys the distribution that we analyzed, we designed the function (11) with opposite monotonicity of (10). We also designed function (12) which has the same monotonicity as function (11). The difference is that the derivative of (12) is larger than (10) in the same independent variable i. Thus, when the same β is used, the q generated in (12) decreases more rapidly, which means that the network can drop more inattentive tokens in the early stage while retaining more in the middle one. Note that all the independent variable symbols in functions have the same meaning

$$q_i = \sin\left(\beta^{(i+1)}\right) \tag{10}$$

$$q_i = \cos\left(\beta^{(i+1)}\right) \tag{11}$$

$$q_i = \frac{\beta^{(i+1)}}{i+1}. \tag{12}$$

IV. EXPERIMENTS AND ANALYSIS

In this section, three HS datasets that are well designed by the experts are first described. The implementation details of the network and environment configuration are introduced in the second part. Extensive experiments are conducted with ablation analysis to demonstrate the performance of our method both quantitatively and qualitatively in the third part. Finally, state-of-the-art methods are introduced to further demonstrate the superiority of our method.

A. Data Description

1) Indian Pines (IP) Data: The IP dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor which covers agricultural areas in northwestern Indiana and consists of 145 × 145 pixels at ground sampling distance of 20 m. After removing 20 noisy and water absorption bands, there remain 200 spectral bands in the range of 400–2500 nm with 10 m spectral resolution. Sixteen land cover classes were labeled in this scene, and the training and testing samples are detailed in the second column of Table III.

2) Salinas Data: The Salinas (SA) dataset was captured by the AVIRIS sensor which consists of 512 × 217 pixels with ground sampling distance of 3.7 m. This dataset has 204 spectral bands after discarding the water absorption bands. All labeled pixels were divided into 16 land cover classes, and the training and testing samples are detailed in the third column of Table III.

3) Kennedy Space Center (KSC) Data: The KSC dataset was captured by Jet Propulsion Laboratory's (JPL) AVIRIS at the KSC, Florida. It consists of 176 spectral bands and 512 × 614 pixels. The labeled pixels contain 13 categories, and the training and testing samples are detailed in the fourth column of Table III.

B. Experimental Setting

1) Evaluation Metrics: Three evaluation metrics including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are used to quantitatively show the superiority of the proposed methods. Moreover, the classification maps are generated to show qualitative comparison intuitively.

2) Experimental Configuration: All the methods have been operated on the same workstation, with i7-8700 CPU and an NVIDIA GeForce GTX 1080Ti graphics processing unit (GPU) with 11 GB RAM. The software environment is implemented by PyTorch with compute unified device architecture (CUDA) 10.1 and cuDNN 7.1. We set the epoch as 600 for all the datasets. The learning rate is 1e-3 for the IP dataset and 1e-4 for the SA and KSC datasets. The Adam optimizer [55] is adopted with a batch size of 64, and the learning rate will be multiplied by a factor of 0.9 per 50 epochs.

C. Parameter Analysis

1) Ablation Study: We conduct three ablation studies to prove our hypothesis and to investigate the effectiveness of different components in the proposed SSPT.

First, we investigate the TR operation without any compensation mechanism. The OA score is listed in Table IV. In detail, the original ViT without any supplementary information yields the lowest OA score. When the neighboring bands and spatial information are added, the OA score shows an upward trend in the limited data volume range, but the performance behaves unstably when the input data volume continues to increase. We deem that too much input data will lead to an increase in redundant features under a limited training sample. By introducing the TR operation, the classification result of ViT + TR is better than its counterparts. Moreover, with the increase of neighboring bands, more valid features are retained because the TR operation can progressively filter redundant features, so the OA scores are relatively stable within a certain range.

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TABLE III
LAND COVER CLASSES OF THREE DATASETS WITH TRAINING AND TESTING SAMPLES

Datasets	**Indian Pines (IP)**			**Salinas (SA)**			**Kennedy Space Center (KSC)**		
Class No.	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Corn Notill	50	1384	Brocoli green weeds 1	50	1959	Scrub	50	297
2	Corn Mintill	50	784	Brocoli green weeds 2	50	3676	Willow swamp	50	193
3	Corn	50	184	Fallow	50	1926	CP hammock	50	206
4	Grass Pasture	50	447	Fallow rough plow	50	1344	Slash pine	50	202
5	Grass Trees	50	697	Fallow smooth	50	2628	Oak/broadleaf	50	111
6	Hay Windrowed	50	439	Stubble	50	3909	Hardwood	50	179
7	Soybean Notill	50	918	Celery	50	2529	Swamp	50	55
8	Soybean Mintill	50	2418	Grapes untrained	50	11221	Graminoid marsh	50	340
9	Soybean Clean	50	564	Soil vinyard develop	50	6153	Spartina marsh	50	470
10	Wheat	50	462	Corn senesced green weeds	50	3228	Cattail marsh	50	354
11	Woods	50	1244	Lettuce romaine 4wk	50	1018	Salt marsh	50	369
12	Buildings Grass Trees Drives	50	330	Lettuce romaine 5wk	50	1877	Mud flats	50	453
13	Stone Steel Towers	50	45	Lettuce romaine 6wk	50	866	Water	50	877
14	Alfalfa	15	39	Lettuce romaine 7wk	50	1020			
15	Grass Pasture Mowed	15	11	Vinyard untrained	50	7218			
16	Oats	15	5	Vinyard vertical trellis	50	1757			
	Total	695	9671	Total	800	53329	Total	650	4561

TABLE IV
ABLATION ANALYSIS OF THE PROPOSED TR METHOD OF SSPT IN TERMS OF OA (%) ON THE IP DATASET

Methods	**Input**	**Group Size** (g)				
		1	3	5	7	9
ViT	pixel-wise	72.36	74.83	75.86	74.67	74.09
ViT+TR		73.12	75.46	76.67	**77.52**	75.95
ViT	patch-wise	80.08	82.28	83.88	83.64	83.65
ViT+TR		81.52	83.77	84.23	84.56	**85.03**

Second, we investigate the ATC mechanism based on the TR operation. This experiment can further prove our hypothesis that in the early stage of feature extraction, the inattentive tokens may contain more redundant features than later ones. As shown in Table V, by plugging the TF strategy for all stages of ViT, the OA scores do not significantly increase, even a slight decrease. For comparison, we empirically pick a set of q [i.e., (1.0, 0.75, 0.5, 0.25, 0.0)] for the different stages as the initial version of the ATC, which will be subsequently generated by a predefined function. Compared with TF, the ATC mechanism can automatically select the relatively effective features from the inattentive token and tends to obtain the best performance.

Third, different q generation functions and hyperparameter β values have been experimented. It should be noted that the functions selected in the experiment are not necessarily optimal; we just choose those that can prove our hypothesis about the distribution of supplementary information contained in inattentive tokens. The functions A, B, and C represent (10)–(12), respectively. The OA value calculated by case 1 of Table II is set as the baseline. According to Fig. 3, half of the results calculated by function B lie below the baseline; the reason is that the distribution of the quantile calculated by function B is opposite to the distribution we assume for the valid information contained in the inattentive tokens. For functions A and C, most of their calculated values lie above the baseline, which further proves the correctness of our hypothesis. In particular, when the β values are 0.7 and 0.5, the OA values of functions A and C are 3% and 2% above the

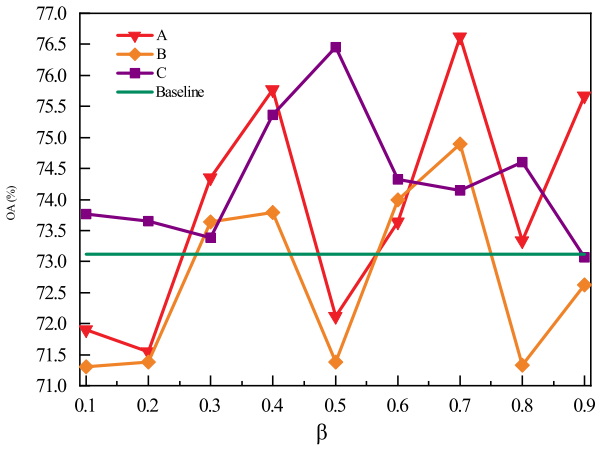

Fig. 3. Effect of quantile generated by different functions on the OA.

baseline, respectively. It should be noted that the functions A and C have the same monotonicity, which determines that the distribution trend of the quantile is the same; the difference is that with the same β and change step size of i, the trend of A changes more slowly as the number of stages increases, which means that compared to C, A will discard more tokens in the middle stage of the network.

2) Parameter Sensitivity Analysis: We conduct experiments on some key parameters in the proposed network as the way to evaluate their impact on the classification performance. Three parameters including patch size (s) of the data cube, keep rate (k) of every encoder stage, and the number of group size (g) have been experimented.

a) Evaluate Patch Size: The patch size determines the amount of spatial information that the network can learn from the original HSI data cube. However, too large a patch size may increase the difference between data and labels, because the label of the center pixel is defined as the label of the patch data. To explore the optimal patch size for the proposed SSPT, we conduct the size of the patch from 1×1 to 13×13. The classification accuracy influenced by different patch sizes is listed in Table VI. We can conclude that the optimal patch

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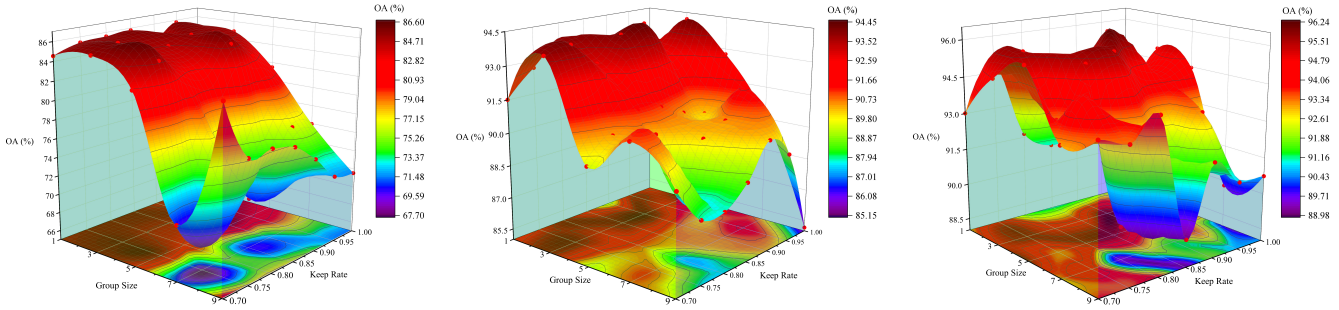
TABLE V
ABLATION ANALYSIS OF THE PROPOSED ATC MECHANISM OF SSPT ON THREE DATASETS

Dataset		IP			SA			KSC		
Metrics		OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Methods	ViT+TF	75.12	81.89	71.83	84.68	92.45	83.02	84.48	80.50	82.65
	ViT+ATC	76.39(↑**1.27**)	83.14(↑**1.25**)	74.79(↑**2.96**)	85.52(↑**0.84**)	92.96(↑**0.51**)	83.93(↑**0.91**)	87.26(↑**2.78**)	83.17(↑**2.67**)	85.74(↑**3.09**)



(a) (b) (c)

Fig. 4. Impact between different group sizes and keep rate for the OA. (a) IP, (b) Salinas, and (c) KSC.

TABLE VI
IMPACT OF DIFFERENT PATCH SIZES IN TERMS OF OA (%) ON THREE DATASETS

Patch Size	1×1	3×3	5×5	7×7	9×9	11×11	13×13
Indian Pines	72.36	80.08	83.16	**84.36**	83.46	82.62	83.73
Salinas	80.86	87.38	89.18	93.30	89.47	**94.03**	93.66
Kennedy Space Center	84.56	80.68	87.92	89.45	87.72	**93.13**	90.74

size is 7 × 7 for the IP dataset and 11 × 11 for the SA and KSC datasets.

b) Mutual Effect of Keep Rate and Group Size: A common conclusion is that the group-wise tokenization method can introduce more detailed sequential properties of the spectra which contain subtle spectral discrepancies. However, this method exponentially increases the redundant information between spectral bands which is determined by group size g. The TR method that we proposed can progressively extract the tokens that are more effective for the classification by the SA mechanism and the keep rate k determines the proportion of tokens that are considered valid features for each encoder block. The amount of effective spectral embeddings involved in the feature extraction stage is decided by the interaction of parameters g and k, and the mutual effect of them is shown in Fig. 4. The group size lies in the range of [1] and [9] with step size 2 and the keep rate lies in [0.7, 1] with step size 0.5. It can be seen that both g and k have an effect on OA, and their relationship is difficult to describe by a linear function. Although the TR method can filter out the redundant features from original data even with large group size, when g is too large (over 5), even if we use a smaller k to reduce the number of tokens, there is still a cliff-like decrease in accuracy. The main reason may be that the diverse semantic information of the sequential image cannot be efficiently extracted by a fixed architecture and limited learnable parameters. However, when g lies in range [1, 5] and k lies in range [0.7, 0.9], the properties of locally convex functions are obtained for three datasets. This means that we can experimentally choose the optimal g and k pairings for different datasets to obtain the best classification performance.

D. Comparison With Other Methods

To fully demonstrate the effectiveness of the SSPT, a series of representative backbones are selected for quantitative comparison.

For the SVM, the radial basis function (RBF) kernel is introduced for classification. The hyperparameters σ and λ of RBF are set after cross-validation in range of $\sigma = [2^{-3}, 2^{-2}, \ldots, 2^4]$ and $\lambda = [10^{-2}, 10^{-1}, \ldots, 10^4]$. For the KNN, the number of nearest neighbors (K) is set to 10. For the 1-D CNN [56], each unit is composed of 1-D conventional kernel with the output size of 128, a batch normalization (BN) layer, and a ReLU activation function. The total network includes five units. Finally, the calculation results of the softmax function provide the basis for the final classification task. For the 2-D CNN [11], in addition to the elements contained in the 1-D unit, the max-pooling layer has been added. Each 2-D unit contains 8, 16, and 32 2-D kernels with the size of 3 × 3, respectively. Finally, two linear layers with the softmax function are used to generate the classification result. For the ViT [26], LocalViT [57], HiT [35], LeViT [38], RvT [39], and SpectralFormer [29], we adopt the author's source code directly and set the parameters as detailed in their paper. For a fair comparison, the proposed SSPT adopts the same encoder number, hidden dimensions of MLP, and head number of MHSA mechanism as the above-mentioned transformer-based methods, if the modification of the network architecture is not necessary.

1) Quantitative Result: Three indices (i.e., OA, AA, and κ) and the classification accuracy for each class of IP, Salinas, and KSC HS datasets are listed in Tables VII–IX, respectively.

In general, conventional classifiers are lower than deep learning methods in three classification metrics. The high classification performance of deep learning methods stems

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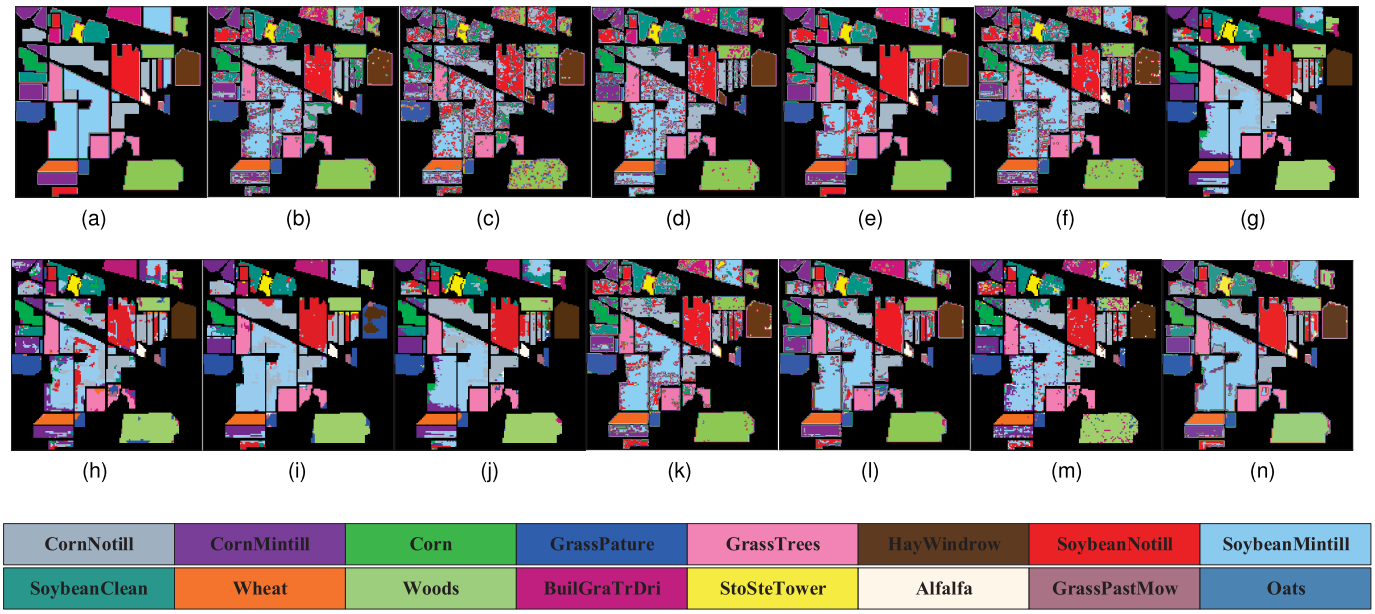

Fig. 5. Ground truth and classification maps obtained by different methods on the IP dataset. (a) Ground truth. (b) SVM. (c) KNN. (d) 1-D-CNN. (e) 2-D-CNN. (f) ViT. (g) LocalViT. (h) RvT. (i) LeViT. (j) HiT. (k) SF-Pixel. (l) SF-Patch. (m) SSPT-Pixel. (n) SSPT-patch.

from the powerful feature extraction capability of the backbone networks. Different backbone networks have different types of features that are of primary concern. For example, 1-D CNN, pixel-wise SF, and SSPT are better modeling the sequential properties of the spectral dimension and 2-D CNN is more concerned with spatial information. We can see from the tables that the classification accuracy of 2-D CNN is higher than backbones that focus only on spectral information. It is worth noting that our proposed pixel-wise SSPT achieves the best classification performance under the situation that relies only on spectral information, and the classification indicators are even close to 2-D CNN. (For example, on the IP dataset, its OA is only 0.28% lower than 2-D CNN.) Other transformer-based methods (i.e., LocalViT, HiT, patch-wise ViT, SF, and SSPT) are considered to be able to extract the joint spatial–spectral features because the spatial information is contained in HSI patches. Except for ViT, all other patch-wise transformer-based methods performed significantly better than 2-D CNN, which may be because the pure ViT is difficult to model local spatial information effectively. Specifically, the proposed patch-wise SSPT, as the pure transformer-based method, achieves the highest classification result than its counterparts. The reason may be that SSPT enables the network to learn effective features more efficiently by discarding redundant information.

To further verify the effectiveness of SSPT, miniGCN [58], FuNet-C [58], MAEST [34], DeepViT [36], and T2T [37] were selected as further methods of comparison. Since the code of the above-mentioned methods is not publicly available, we fail to achieve their reported accuracy when retrained on our platform. Therefore, the same approach of dividing the training and testing samples is deployed on SSPT for a fair comparison. Specifically, miniGCN, FuNet-C, and MAEST used the same sampling strategy as us, and we adopt 10% of labeled pixels of the IP dataset as training samples to directly compare with T2T and DeepViT methods.

From the left part of Table X, for these methods with the same data sampling strategy, the patch-based SSPT obtains the best general results among the studied methods, because it reduces the effect of redundant information. For the right part, when we adopt 10% of data as training samples, the proposed SSPT is much better than the comparison methods. Specifically, our method has increased by 14.53% and 18.37% in terms of OA when compared to DeepViT and T2T, respectively.

2) Visual Evaluation: To visualize the differences in the features learned by different backbone networks and to demonstrate the superiority of our proposed method, the classification maps obtained by different methods are shown in Figs. 5–7 for IP, SA, and KSC datasets, respectively. We enlarge the classification results of some representative categories in the KSC dataset to show the differences between the different methods more clearly in Fig. 8.

It is obvious that the classification results of the traditional classifier are coarse, and a large number of pixels are misclassified. In contrast, the overall classification results of deep learning methods are relatively smooth. Although CNN-based methods and pixel-wise transformer-based methods show similar results in terms of classification indicators, owing to powerful spatial–contextual features fitting ability, the classification maps obtained from the CNN-based approach behave more smoothly at the spatial dimension (especially for the SA dataset). With the addition of spatial information, patch-wise transformer-based methods perform more competitively than CNN-based methods. Benefitting from the special tokenization, the patch-wise SF and SSPT learn more spatial–spectral signatures with limited training samples which helps to achieve fine-grained classification. According to the classification maps, SF and SSPT have higher classification accuracy than other methods when dealing with very few samples and boundary pixels. Note that our proposed SSPT

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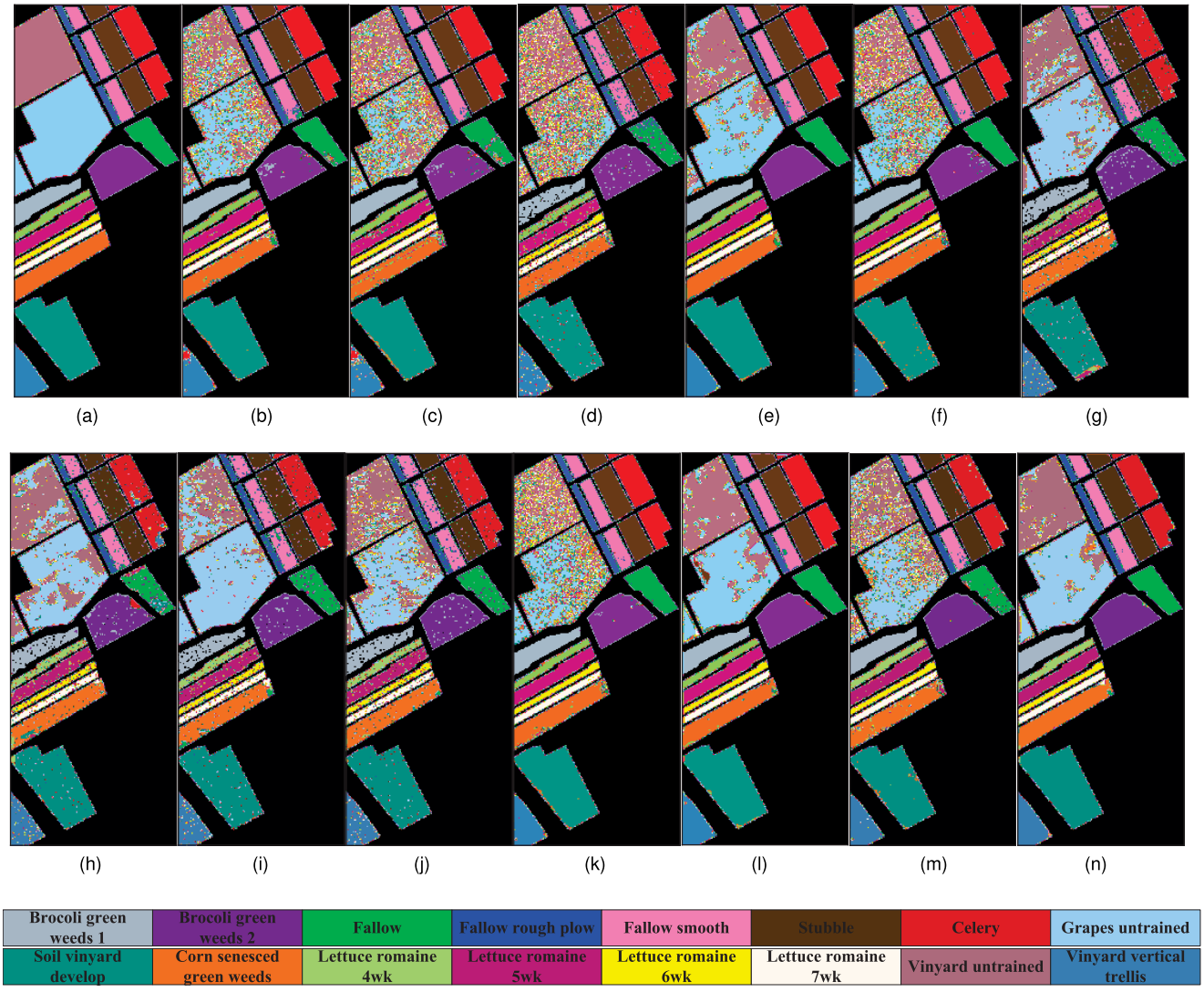

Fig. 6. Ground truth and classification maps obtained by different methods on the Salinas dataset. (a) Ground truth. (b) SVM. (c) KNN. (d) 1-D-CNN. (e) 2-D-CNN. (f) ViT. (g) LocalViT. (h) RvT. (i) LeViT. (j) HiT. (k) SF-Pixel. (l) SF-Patch. (m) SSPT-Pixel. (n) SSPT-patch.

TABLE VII

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE IP DATASET. THE BEST ONE IS SHOWN IN BOLD

NO.	SVM	KNN	1-D CNN	2-D CNN	ViT	LocalViT	RvT	LeViT	HiT	SF		SSPT	
										pixel-wise	patch-wise	pixel-wise	patch-wise
1	58.74	43.64	73.41	81.35	49.85	78.31	70.80	78.66	69.29	66.83	68.64	70.08	**82.65**
2	64.03	48.85	53.95	85.84	62.50	83.69	77.67	**90.52**	84.44	66.96	87.50	76.62	88.77
3	78.80	76.08	90.21	90.76	83.69	97.01	95.65	94.44	95.65	92.39	85.32	87.18	**97.28**
4	88.36	80.76	25.95	89.71	88.81	93.36	92.84	92.96	91.50	92.84	**95.74**	89.54	94.40
5	82.35	77.18	94.11	**99.71**	86.65	89.96	79.34	91.56	94.84	84.79	85.79	90.50	94.83
6	95.89	94.98	99.08	97.04	92.25	**100.00**	**100.00**	39.26	**100.00**	94.30	97.49	96.93	96.12
7	77.66	67.53	68.30	**95.75**	80.39	73.76	66.44	91.32	77.02	88.01	85.29	83.88	89.32
8	60.09	46.36	58.27	48.09	71.67	57.26	55.58	82.45	71.05	71.50	76.88	71.96	**82.96**
9	57.97	39.71	75.31	70.56	46.09	75.90	73.04	73.61	68.44	72.16	**78.90**	74.92	76.06
10	97.53	98.76	99.38	**100.00**	98.76	99.53	98.14	**100.00**	**100.00**	98.76	99.38	98.58	99.38
11	86.65	74.43	89.87	**98.55**	90.67	91.42	90.27	95.74	94.05	93.01	93.24	80.76	92.68
12	69.69	18.78	75.75	90.60	48.48	96.32	95.45	**97.36**	96.67	59.69	81.81	80.03	86.96
13	86.66	91.11	93.33	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
14	69.23	53.84	1.13	79.48	94.87	96.30	87.17	98.14	82.05	87.17	71.79	81.48	**100.00**
15	81.82	81.81	1.05	**100.00**	90.90	**100.00**	**100.00**	**100.00**	**100.00**	90.91	**100.00**	96.15	**100.00**
16	**100.00**	60.00	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	95.10	**100.00**
OA(%)	71.23	58.21	70.83	79.69	72.50	79.28	74.76	85.16	81.30	78.50	82.87	79.41	**87.68**
AA(%)	78.84	65.86	68.57	89.22	79.55	89.55	86.48	89.13	89.86	84.96	87.99	85.85	**92.59**
$\kappa \times 100$	67.51	53.11	66.83	77.15	68.50	76.71	71.47	83.12	78.81	75.55	80.45	76.81	**85.95**

can reduce the impact of redundant information on the model learning ability in an end-to-end manner that enables the network to learn spatial–spectral information more efficiently and ultimately obtain more robust classification results.

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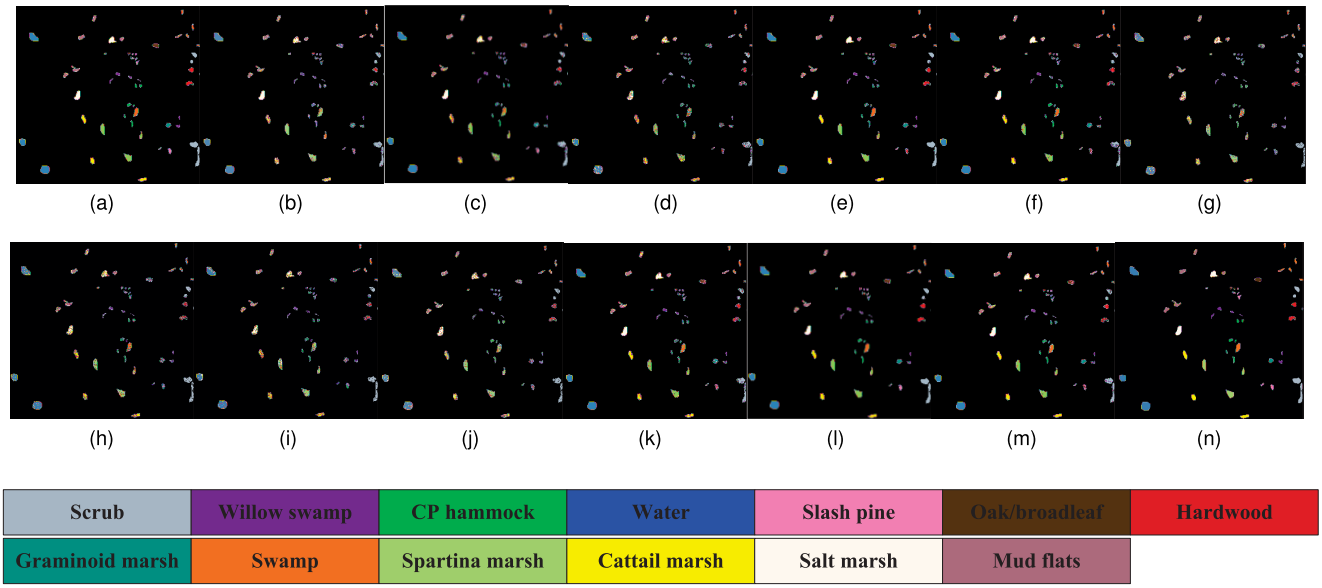

Fig. 7. Ground truth and classification maps obtained by different methods on the KSC dataset. (a) Ground truth. (b) SVM. (c) KNN. (d) 1-D-CNN. (e) 2-D-CNN. (f) ViT. (g) LocalViT. (h) RvT. (i) LeViT. (j) HiT. (k) SF-Pixel. (l) SF-Patch. (m) SSPT-Pixel. (n) SSPT-patch.

TABLE VIII

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE SALINAS DATASET. THE BEST ONE IS SHOWN IN BOLD

NO.	SVM	KNN	1-D CNN	2-D CNN	ViT	LocalViT	RvT	LeViT	HiT	SF		SSPT	
										pixel-wise	patch-wise	pixel-wise	patch-wise
1	97.36	96.27	99.54	99.89	99.48	**100.00**	92.70	**100.00**	99.49	98.92	94.99	99.28	96.73
2	83.90	89.82	99.81	99.86	97.47	98.09	95.40	99.86	**99.97**	98.74	98.09	99.59	99.40
3	98.66	98.80	92.52	99.84	99.16	98.85	78.71	98.07	**100.00**	99.06	99.89	95.84	99.84
4	97.14	93.30	99.77	**99.92**	95.90	99.33	99.85	97.09	98.73	99.85	98.21	99.11	99.26
5	99.33	99.21	96.15	97.41	97.22	99.42	98.21	97.94	**99.58**	97.37	98.09	96.68	99.51
6	99.23	98.75	99.82	99.89	95.59	96.62	**100.00**	99.97	99.97	99.69	99.31	95.47	99.10
7	70.93	67.73	**99.91**	98.10	99.34	98.67	92.77	98.18	98.53	99.46	98.58	97.76	96.71
8	**98.60**	97.28	55.61	58.96	69.01	84.12	76.59	95.56	76.15	70.66	78.13	66.03	87.46
9	83.33	83.64	99.70	99.05	96.44	95.14	97.82	99.43	**99.71**	97.62	98.19	97.61	99.41
10	90.37	88.51	90.73	90.21	86.77	94.64	91.69	89.90	**97.18**	88.63	91.41	88.28	94.83
11	99.84	94.35	92.43	99.11	95.77	96.56	97.93	**100.00**	99.21	93.94	97.05	88.01	99.31
12	98.49	97.69	99.78	98.98	**99.89**	98.93	97.81	99.78	99.47	99.84	97.76	98.45	99.84
13	90.09	91.17	99.19	**100.00**	98.03	99.53	98.15	**100.00**	97.34	98.96	98.15	98.04	98.26
14	48.29	60.31	91.96	99.11	91.56	**100.00**	99.01	55.79	97.84	93.92	98.62	96.56	97.84
15	91.74	89.81	73.81	94.69	63.92	79.83	71.72	**100.00**	78.63	60.66	88.88	73.13	89.42
16	98.31	96.98	98.01	98.74	97.43	**100.00**	98.86	**100.00**	99.94	97.55	98.12	98.41	99.14
OA(%)	83.94	84.34	85.62	89.55	86.41	92.27	88.37	92.05	91.59	86.86	92.31	86.93	**94.94**
AA(%)	90.35	90.22	93.05	95.86	92.88	96.19	92.95	95.73	96.36	93.43	95.85	93.02	**97.26**
$\kappa \times 100$	82.09	82.58	84.06	88.44	84.91	91.40	87.05	91.10	90.65	85.38	91.46	85.49	**94.37**

TABLE IX

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE KSC DATASET. THE BEST ONE IS SHOWN IN BOLD

NO.	SVM	KNN	1-D CNN	2-D CNN	ViT	LocalViT	RvT	LeViT	HiT	SF		SSPT	
										pixel-wise	patch-wise	pixel-wise	patch-wise
1	90.43	87.34	90.58	97.81	82.98	97.89	95.78	97.04	**99.01**	83.26	98.45	89.17	96.20
2	84.45	80.31	83.41	95.30	74.09	**98.96**	95.33	97.92	97.41	85.49	92.27	83.93	**98.96**
3	29.61	83.98	97.57	54.42	90.77	97.08	85.92	**100.00**	75.24	66.50	87.86	80.09	94.17
4	49.50	52.47	00.00	45.04	54.95	**87.12**	84.65	63.86	41.10	41.08	73.76	52.47	72.28
5	25.22	38.73	61.26	83.96	60.36	86.48	89.18	**95.49**	86.48	50.45	72.07	54.05	92.79
6	43.01	53.63	70.94	80.90	56.98	83.79	**93.85**	88.82	82.12	53.07	92.17	62.56	91.62
7	90.90	90.91	94.54	**100.00**	94.54	**100.00**	**100.00**	**100.00**	**100.00**	83.63	**100.00**	89.09	**100.00**
8	43.83	72.70	90.26	82.04	85.82	92.91	94.75	66.40	78.21	78.74	93.17	86.35	**98.16**
9	81.06	81.70	86.80	87.14	95.74	96.17	**98.29**	91.70	97.23	87.23	**98.29**	94.25	90.21
10	63.27	71.18	90.39	96.52	83.05	94.63	99.15	**100.00**	85.59	83.05	99.15	96.32	99.15
11	89.43	89.97	98.37	**100.00**	97.83	**100.00**	91.05	**100.00**	92.68	94.57	99.72	95.39	99.73
12	82.78	79.47	91.61	92.81	85.43	89.84	99.55	99.13	99.11	84.76	99.77	84.98	**100.00**
13	**100.00**	99.88	98.06	**100.00**	99.88	**100.00**	99.77	**100.00**	**100.00**	99.65	**100.00**	99.77	99.77
OA(%)	76.21	81.67	86.91	89.79	86.54	95.55	95.83	93.55	91.08	82.99	95.88	87.99	**96.08**
AA(%)	67.19	75.56	81.07	85.84	81.73	94.22	94.41	92.34	87.25	76.27	92.82	82.91	**94.85**
$\kappa \times 100$	73.28	79.44	85.33	88.59	84.93	95.00	95.33	92.76	89.98	80.95	95.37	86.52	**95.60**

3) Computational Complexity Analysis: Finally, three metrics (learnable parameters, inference speed, and MACs) were used to show the computational complexity of various methods, where the inference speed represents the time used by the network to process an image patch. To minimize the impact of random initialization parameters and GPU warm-up,

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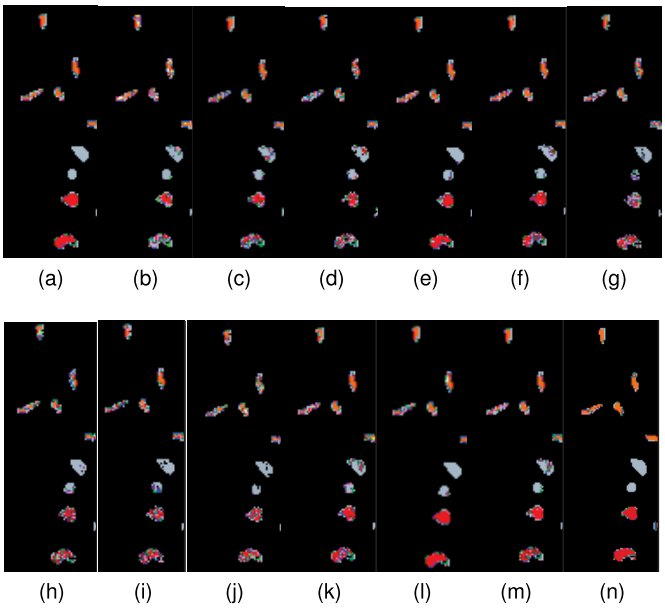

Fig. 8. Local ground truth and classification maps obtained by different methods on the KSC dataset. (a) Ground truth. (b) SVM. (c) KNN. (d) 1-D-CNN. (e) 2-D-CNN. (f) ViT. (g) LocalViT. (h) RvT. (i) LeViT. (j) HiT. (k) SF-Pixel. (l) SF-Patch. (m) SSPT-Pixel. (n) SSPT-patch.

TABLE X
CLASSIFICATION ACCURACY OF DIFFERENT METHODS WITH DIFFERENT SAMPLING STRATEGIES ON THE IP DATASET

Methods	minGCN	FuNet-C	MAEST	SSPT	DeepViT	T2T	SSPT
OA	75.11	79.89	84.15	87.68	75.01	71.17	89.54
AA	78.03	89.35	90.97	92.59	-	-	81.31
kappa	71.64	77.16	82.00	85.95	71.65	67.29	88.07

TABLE XI
COMPUTATIONAL COMPLEXITY OF SELECTED METHODS

Model	Prameters (M)	Inference Speed (ms)	MACs (G)
1-D CN	0.03	-	0.001
2-D CNN	1.67	0.17 (0.04)	0.03
ViT	3.77	2.52 (0.10)	0.73
LocalViT	31.85	5.48 (0.41)	0.78
RvT	14.47	7.26 (0.43)	0.03
LeViT	16.95	16.23 (0.51)	0.89
HiT	56.81	24.51 (0.84)	0.64
SF	0.35	2.97 (0.45)	0.03
DeepViT	52.21	-	1.36
T2T	730.18	11.05 (0.54)	2.97
SSPT	0.33	5.29 (0.52)	0.04

we counted the average time for 100 sets of image patches, with the standard deviation of the time used in parentheses. Since the 1-D CNN uses pixel-wise input and the details of DeepViT are not clear, we do not report their inference speed in the experiments. It should be noted that inference times reported in this article may vary based on the platform and the coding strategies used when implementing these methods; therefore, the results may be biased. The results are shown in Table XI.

In general, the CNN-based methods outperform the transformer-based methods in terms of computational complexity and inference time. This is mainly because the implementation of the SA operation of transformer relies on fully connected layers which have a larger number of parameters than convolutional networks at the same feature scale. When considering the transformer-based methods, SSPT shows significant advantages in terms of number of parameters and MACs (except SF) while producing the best classification performances. In particular, the number of parameters is lower than 2-D CNN, but the classification performance is much higher than it. However, it can be seen that the inference time of SSPT is not the best. The reason may be that the quantile generation operation and ATC operation are time-consuming. Notably, it still takes less time than most transformer-based methods. Taking all aspects into account, the results demonstrate the efficient feature representation ability of SSPT.

V. CONCLUSION

Since the transformer shows a powerful ability in capturing long-term dependences by its global receptive field of SA mechanism, many transformer-based backbone networks have been used to model the long-range interspectra dependences for HSIs classification. However, the existing transformer-based backbones inevitably suffer from the Hughes phenomenon, especially with special group-wise tokenization. To this end, a new lightweight transformer-based network named SSPT is proposed, which fully exploits the characteristics of the transformer while mitigating the impact of redundant features on the backbone network in an end-to-end manner. In specific, the TR method can progressively generate effective HSI representation and discard tokens containing redundant features by SA mechanism, which significantly reduces the computational burden of the network while improving the classification accuracy. In addition, we experimentally verify the hypothesis that the relative valid information in redundant tokens is distributed differently at different stages, and propose an ATC mechanism to collect supplementary features that can further improve the classification accuracy. Meanwhile, the methods we have adopted in SSPT can be easily embedded in other transformer-based backbones for HSI downstream tasks.

In the future, we will explore a more efficient way of tokenization, which can increase the amount of effective information while reducing redundant information both in the spatial and spectral dimensions. We also attempt to efficiently integrate CNN and transformer to integrate the merits of both backbone networks while keeping the network as lightweight as possible. Furthermore, the hypothesis of the distribution of effective features in different stages of the transformer is an important factor to improve the network performance and reduce the computational burden, which deserves to be further explored.

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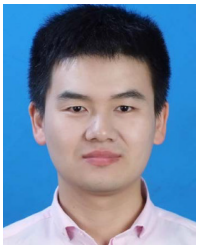

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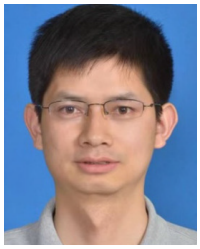

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